

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** Script-Controlled ACL – Restrict Record Access Based on Field Value
- **Team Members:** [Member Name 1] – [Role], [Member Name 2] – [Role], [Member Name 3] – [Role]

2. Project Overview

- **Purpose:** This lab demonstrates how to implement a script-controlled Access Control List (ACL) to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.
- **Features:** Script-controlled READ, WRITE, CREATE and DELETE ACLs enforce record-level security. User roles and record field values are evaluated together to make access decisions. ACLs protect sensitive data from unauthorized viewing, modification, creation and deletion across forms, lists and Service Portal interfaces.

3. Architecture

- **Frontend:** ServiceNow instance interface – Application Navigator, the Institution Details list (u_institution_details.list) and record forms, where users view records according to their roles.
- **Backend:** Access Control (ACL) rules under System Security, created with the security_admin role elevated. The READ ACL uses an advanced script that checks roles with gs.hasRole(); CREATE, WRITE and DELETE ACLs use required roles (bb2, bb3, bb4).
- **Database:** Custom table Institution Details (u_institution_details), Extends for: false, with the following fields:
 - o Student Roll Number – Auto Number
 - o Student Name – Reference (User)
 - o Faculty Name – Reference (User)
 - o Branch – Choice (ECE, EEE, CSE)
 - o Email – String
 - o Phone Number – String
 - o Description – Multi String

4. Setup Instructions

- **Prerequisites:** ServiceNow instance with admin access; security_admin role elevation (required to create ACLs).
- **Installation:** Step-by-step guide (Phases 1–3):
 - o **Phase 1 – Users Creation:** Log in to the ServiceNow instance with admin access. Create a test user.
 - Navigate to User Administration → Users and click New.
 - User ID: EEE User; First name: EEE; Last name: User; Email: eeeuser@gmail.com.

- Save the record.
- o **Phase 2 – Roles Creation:** Navigate to User Administration → Roles.
 - Click New; Name: bb1; Description: brief role purpose; click Submit.
 - Create the other roles bb2, bb3 and bb4 by following the same procedure.
 - Assign all the roles to the EEE User.
- o **Phase 3 – Creation of Table:** In the Application Navigator, search for System Definition → Tables.
 - Click New; Label: Institution Details; Name: u_institution_details; Extends for: false; click Submit.
 - Add the fields listed under Database (Section 3) and save the table.
 - Create multiple records in the Student Records table with different Branch values (ECE, EEE, CSE).

5. Folder Structure

- **Client:** User Administration → Users (EEE User); User Administration → Roles (bb1, bb2, bb3, bb4); System Definition → Tables (u_institution_details); u_institution_details.list (Student Records list).
- **Server:** System Security → Access Control (ACL) – Read, Create, Write and Delete ACLs on u_institution_details.

6. Running the Application

- After the ACLs are created, verify access as follows:
 - o Impersonate the EEE user and open the Student Records list.
 - o Copy the table name u_institution_details.list/LIST and paste it in the All menu.
 - o Impersonate Admin and open the Student Records list.

7. API Documentation

- ACLs on the table u_institution_details (Type: record, Active: true):
 - o **Read:** Requires role bb1; data condition Branch is EEE; Advanced: true with script (see Section 8).
 - o **Create:** Requires role bb2; no data condition.
 - o **Write:** Requires role bb3; no data condition.
 - o **Delete:** Requires role bb4; no data condition.

8. Authentication

- Authorization is handled through roles (bb1–bb4) assigned to the EEE User and record-level ACLs; the admin user always has full access. ACLs are created after elevating the security_admin role.
 - o **Phase 4 – ACL Configuration (Read ACL):** Search for System Security → Access Control (ACL), elevate “Security_admin” and click New.
 - Type: record; Operation: read; Name: u_institution_details; Active: true; Advanced: true.
 - In the Requires role related list, add bb1 (custom role).

- Add Data condition: Branch is EEE.
- In the Script field, add the following script, then click Submit:

```
(function () {
  // Allow admin users full access
  if (gs.hasRole('admin')) {
    return true;
  }

  // Allow only EEE branch users to see EEE records
  if (gs.hasRole('bb1')) {
    return true;
  }

  // Deny access for all others
  return false;
})();
```

o **Phase 5 – Creation of ACL-Create:** Click New.

- Type: record; Operation: create; Name: u_institution_details; Active: true.
- In the Requires role related list, add bb2 (custom role). No data condition. Save the ACL record.

o **Phase 6 – Creation of ACL-Write:** Click New.

- Type: record; Operation: Write; Name: u_institution_details; Active: true.
- In the Requires role related list, add bb3 (custom role). No data condition. Save the ACL record.

o **Phase 7 – Creation of ACL-Delete:** Click New.

- Type: record; Operation: Delete; Name: u_institution_details; Active: true.
- In the Requires role related list, add bb4 (custom role). No data condition. Save the ACL record.

9. User Interface

The screenshot shows the ServiceNow 'Users' page. At the top, there's a navigation bar with 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a 'Users' button. Below this is a search bar and a table of users. The table has columns: User ID, Name, Email, Active, Created, and Updated. The first row is highlighted for 'EEE User'.

User ID	Name	Email	Active	Created	Updated
EEE User	EEE User	eeeuser@gmail.com	true	2026-09-26 04:37:33	2026-09-26 04:37:33
		example.com	true	2012-02-17 19:04:52	2026-09-01 15:41:42
		example.com	true	2013-07-23 17:15:54	2026-09-01 15:41:46
		example.com	true	2012-02-17 19:04:50	2026-09-01 15:41:35
		example.com	true	2012-02-17 19:04:49	2026-09-01 15:41:43
		example.com	true	2012-02-17 19:04:52	2026-09-01 15:41:37
		example.com	true	2012-02-17 19:04:52	2026-09-01 15:41:46
		example.com	true	2012-02-17 19:04:53	2026-09-01 15:41:47
		example.com	true	2012-02-17 19:04:51	2026-09-01 15:41:36
		example.com	true	2012-02-17 19:04:52	2026-09-01 15:41:43
		example.com	true	2012-02-17 19:04:53	2026-09-01 15:41:46
		example.com	true	2012-02-17 19:04:52	2026-09-01 15:41:46
		example.com	true	2012-02-17 19:04:50	2026-09-01 15:41:35
		example.com	true	2012-02-17 19:04:50	2026-09-01 15:41:48
		example.com	true	2012-02-17 19:04:52	2026-09-01 15:41:37

Below the table, there's a detailed view of the 'EEE User'. It includes fields for User ID, First name, Last name, Title, Department, Password, Password needs reset, Locked out, Active, Identity type, Email, Language, Notification, Calendar integration, Time zone, Date format, Business phone, Mobile phone, and Photo. The 'Open Record' button is highlighted.

Screenshot – User Creation

servicenow All Favorites History Workspaces Roles Search

Roles Name Search Actions on selected rows... New

All > Name >= bb1

Name	Description	Elevated privilege
Search	Search	Search
bb1		false
bb2		false
bb3		false
bb4		false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate	false

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Screenshot – Role Creation

servicenow All Favorites History Admin Table - Institution Details Search

Table Institution Details Delete Update Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Institution Details Application Global

* Name u_institution_details Remote Table

Is Data Interface

Columns Controls Application Access

Table Columns for text Search 1 to 13 of 13 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Branch	Choice	(empty)	40		false
Faculty Name	Reference	User	32		false
Phone Number	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Student Roll Number	String	(empty)	40		false

Screenshot – Table Creation

The screenshot shows the 'Table - Institution Details' configuration page in ServiceNow. It displays a table of 'Dictionary Entries' with the following data:

Column label	Type	Reference	Max length	Default value	Display
Branch	Choice	(empty)	40		false
Faculty Name	Reference	User	32		false
Phone Number	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Student Roll Number	String	(empty)	40		false
Created by	String	(empty)	40		false
Student Name	Reference	User	32		false
Created	Date/Time	(empty)	40		false
Email	String	(empty)	40		false
Description	String	(empty)	40		false
Updated by	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updates	Integer	(empty)	40		false

Screenshot – Table Fields

10. Testing

- Testing is done by impersonating users and opening the Student Records list. Verify:
 - User with bb1 role can view only EEE branch records.
 - User without the role cannot view any records.
 - Admin user can view all records regardless of branch.

11. Screenshots or Demo

The screenshot shows the 'Access Control - u_institution_details' configuration page. The configuration is as follows:

- Type: record
- Operation: read
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- Name: u_institution_details
- Description: Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.
- Application: Global
- Active: ☒
- Advanced: ☒

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role: 1 to 1 of 1

Screenshot – ACL Configuration

servicenow All Favorites History : Access Control - u_institution_details Search

Access Control u_institution_details

Type: record Application: Global

Operation: create Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

Name: u_institution_details

Description: Allow create for records in u_institution_details, for users with role bb2.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

1 to 1 of 1

Screenshot – ACL Create Configuration

servicenow All Favorites History : Access Control - u_institution_details Search

Access Control u_institution_details

Type: record Application: Global

Operation: write Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

Name: u_institution_details

Description: Allow write for records in u_institution_details, for users with role bb3.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

1 to 1 of 1

Screenshot – ACL Write Configuration

The screenshot shows the ServiceNow ACL configuration page for 'u_institution_details'. The configuration includes the following fields:

- Type: record
- Operation: delete
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Advanced: ☐
- Decision Type: Allow If
- Protection policy: -- None --
- Name: u_institution_details
- Description: Allow delete for records in u_institution_details, for users with role bb4.

Below the configuration fields is a 'Conditions' section with a blue box containing the following text:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

At the bottom, there is a 'Requires role' field and a pagination indicator showing '1 to 1 of 1'.

Screenshot – ACL Delete Configuration

The screenshot shows the ServiceNow 'Institution Details' table. The table has the following columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. The table contains three rows of data:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200
ECE			Abraham Lincoln		Abel Tuter	STU100
CSE			Andrew Och		Amos Linnan	STU300

At the bottom of the table, there is a pagination indicator showing '1 to 3 of 3'.

Screenshot – Final Record Access Result

12. Known Issues

- No known issues are mentioned in the Phase-wise document.

13. Future Enhancements

- Apply the same script-controlled ACL approach to the other branches (ECE, CSE) and to Service Portal interfaces.